

Dr. Navneet

(Academic name: Navneet Redhu)

School of Mathematical Sciences, NISER Bhubaneswar, Office Room No. 311, Bhubaneswar, Odisha, India

Ph: +91-8901382830

navneet@niser.ac.in | navneetredhu.iiti@gmail.com

Google Scholar: [link](#) | MathSciNet: [link](#)

Professional Summary

Mathematician specializing in applied harmonic analysis and frame theory on LCA groups, with a current research focus on phase retrieval and dynamical sampling for operator-generated frames. Experienced in both pure construction problems (Parseval frames, cyclic frames) and their numerical implementation using MATLAB. Seeking postdoctoral or faculty research positions to extend this work in collaboration with groups working on signal and information processing.

Research Interests

- Applied Harmonic Analysis
- Frame Theory on LCA groups
- Wavelets and Dynamical Sampling
- Phase Retrieval
- Dynamical Frames over Local Fields and Their Applications to Erasures
- Construction of Weaving Information Packets

Experience

National Institute of Science Education and Research (NISER), Bhubaneswar, Odisha, India 2026

- Postdoctoral Fellow, September 7, 2026 – Present

IILM University, Gurgaon, Haryana, India 2026

- Assistant Professor, July 1, 2026 – September 1, 2026

Institute of Mathematics for Industry (IMI), Kyushu University, Fukuoka, Japan 2026

- Visiting Research Fellow, May 5, 2026 – May 18, 2026

Education

Indian Institute of Technology Indore, Simrol, Madhya Pradesh, India 2026

- Ph.D. in Mathematics (Specialization: Harmonic Analysis, CPI: 8.2 in Coursework).
Thesis Title: Characterization And Explicit Construction Of Pairwise Orthogonal Parseval Frames On LCA Groups (under the supervision of Professor Niraj Kumar Shukla).

Indian Institute of Technology Guwahati, India 2019

- M.Sc in Mathematics & Computing (CGPA: 6.92/10).

Government P. G. College, Jind, Haryana, India 2016

- B.Sc (Pass Course) in Mathematics, Physics, and Statistics (Percentage: 75.41).

Awards and Achievements

- Joint Research Project (Short term Researcher) at IMI, Kyushu University, Japan, May 5 – May 18, 2026
- Institute Travel Grant, IIT Indore for 12th International Conference on Harmonic Analysis & PDEs, Spain, 2025.
- International Travel Grant, ANRF (Govt. of India) for Strobl24 More on Harmonic Analysis, Austria, 2024.

Qualifying Examinations

- Qualified GATE 2020 – **AIR-64**.
- Qualified CSIR-UGC NET (LS) June 2019 – **AIR-21**.
- Qualified CSIR-UGC NET (JRF) December 2019 – **AIR-131**.
- Qualified IIT JAM 2017 – **AIR-179**.

Journal Publications

1. **Navneet Redhu**, Anupam Gumber, Niraj K. Shukla (2025), *Construction of pairwise orthogonal Parseval frames generated by filters on LCA groups*, **Applied and Computational Harmonic Analysis**, **74**, paper No. 101708, 27pp. <https://doi.org/10.1016/j.acha.2024.101708>
2. Ole Christensen, **Navneet Redhu**, Niraj K. Shukla (2026), *Cyclic frames in finite-dimensional Hilbert spaces*, **Linear Algebra and its Applications**, **728**, 63–81. <https://doi.org/10.1016/j.laa.2025.08.016>
3. **Navneet Redhu**, Anupam Gumber, Niraj K. Shukla *The Geometry Sampling on unions of Co-compact subgroups*, Under Review, **Journal of Mathematical Analysis and Applications**
4. **Navneet Redhu**, Anupam Gumber, Hartmut Führ, Niraj K. Shukla *Characterization and Construction of Pairwise Orthogonal Parseval Frames with Applications to Sampling*, Under Review, **Bulletin des Sciences Mathématiques**

Manuscripts in Preparation

5. **Navneet Redhu**, Ole Christensen, Niraj K. Shukla *Cyclic Frames over Finite Fields*.
6. **Navneet Redhu**, Lokendra Kumar, Niraj K. Shukla *Dynamical Sampling with Translation Invariant, Convolution and Fourier Multiplier Operators*.
7. **Navneet Redhu**, Royko Oshi-Tomiyashu *Applications of frame theory to diffraction data*.

Invited Talk

- *Introductions to wavelets frames and their applications*, La Trobe seminar at IMI, Kyushu University, Japan, April 2026
- *Wavelet Frame Theory and Shearlets in Data Science and Artificial Intelligence*, Mehta Family School of Data Science and Artificial Intelligence, Indian Institute of Technology (IIT) Roorkee, India.

Talks and Poster Presentations

- Talk in 19th International Discussion Meeting in Harmonic Analysis, IISER Mohali, India, **DMHA 2025** (December 16–19, 2025).
- Talk in National Conference on Mathematical Analysis and Applications, IIT (ISM) Dhanbad, India, **NCMAA 2025** (September 4–6, 2025).

- Poster at 12th International Conference on Harmonic Analysis and Partial Differential Equations, El Escorial, Spain, [ICMAT 2025](#) (June 9–13, 2025).
- Poster at International Conference Strobl24, Strobl, Austria, [Strobl24](#) (June 9–15, 2024).
- Poster at Symposium organized by the Department of Mathematics, IIT Indore, India, [IIT Indore](#) (January 11, 2023).

Workshops and Conferences Attended

- Workshop on Harmonic Analysis: Fourier Multipliers and Related Topics, HRI, Prayagraj, India (January 19 – February 2, 2024).
- Conference – 18th International Discussion Meeting in Harmonic Analysis (DMHA), Organized by Department of Mathematics and Computing, IIT Guwahati, India (December 18–21, 2023).
- International Workshop on Representation Theory and Harmonic Analysis, Organized by Department of Mathematics and Computing, IIT Guwahati, India (December 11–16, 2023).
- Online short-term course on Sampling Theory and its Applications: Signal, Image Processing and Data Science, IIT Indore (January 24–29, 2022) at IIT Indore.

Accepted Conference Presentations

- Oral presentation, 3rd International Workshop on Harmonic Analysis and Operator Theory, Institute of Mathematics of the Romanian Academy, Bucharest, Romania, June 9–12, 2026.
- Poster presentation, at International Conference Strobl26, Strobl, Austria, June 2026.

Teaching Experience

- Teaching Assistant, Linear Algebra and ODE-I, IIT Indore (2021–2025): Conducted tutorials, graded assignments, and mentored undergraduate students.
- Teaching Assistant, Calculus, IIT Indore (2021, 2022, 2024): Assisted with course instruction and evaluation.
- Teaching Assistant, Numerical Methods, IIT Indore (2023): Supported lab sessions and computational assignments.

Technical Skills

- Programming Languages: C, C++
- Computational Tools: MATLAB, Python, $\text{\LaTeX}$
- Operating Systems: Windows, Linux
- MS Office (Word, Excel, PowerPoint)

References

1. Prof. Dr. Niraj Kumar Shukla
Professor, Department of Mathematics
Indian Institute of Technology Indore
Indore – 453552, Madhya Pradesh, India
Email: nirajshukla@iiti.ac.in
2. Prof. Dr. Ole Christensen
Professor, Department of Applied Mathematics and Computer Science
Technical University of Denmark
Lyngby, Denmark
Email: ochr@dtu.dk